

PROJECT REQUIREMENTS DOCUMENT

Project Name	Ecosystem Discovery Application
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Functional Requirements
The system shall display a web-based interactive freshwater ecosystem scene representing a single Saskatchewan-inspired environment, including sky, shoreline, water surface, and underwater zones.
The system shall display animated animal sprites placed within ecologically appropriate zones of the ecosystem scene.
The system shall implement a hint-based discovery system where users select mystery cards representing hidden animals.
The system shall display progressively revealed hints to guide the user toward identifying the selected animal.
The system shall allow users to click animals in the ecosystem scene to attempt identification based on the provided hints.
The system shall provide visual feedback for user actions, including: • Correct selection (card flip, success indicator, sticker unlock) • Incorrect selection (animation feedback and additional hints)
The system shall implement a countdown timer (e.g., 120 seconds) to limit gameplay duration and increase engagement.
The system shall provide a sticker-book collection interface that displays: • Successfully identified animals as unlocked stickers • Unsolved or failed animals as locked placeholders
The system shall display the user's overall progress (e.g., number of animals correctly identified).
The system shall provide start and reset functionality to allow users to begin and replay the game.
The system shall include a bonus interaction where users can gain additional time through environmental actions (e.g., dragging trash to a bin).

Technical/Performance Requirements

The application shall follow the Model–View–Controller (MVC) architectural pattern using modular JavaScript (ES6 modules).
The application shall be built using HTML5, CSS3, and JavaScript as core front-end technologies.
The system shall use a backend database (Supabase) to dynamically fetch animal and hint data via REST API.
The system shall manage runtime game state using a centralized state module.
The application shall be compatible with modern web browsers (e.g., Chrome, Edge, Firefox).
All source code shall be maintained in a GitHub repository organized according to MVC principles.
Task tracking shall be managed using GitHub Projects (Kanban board).